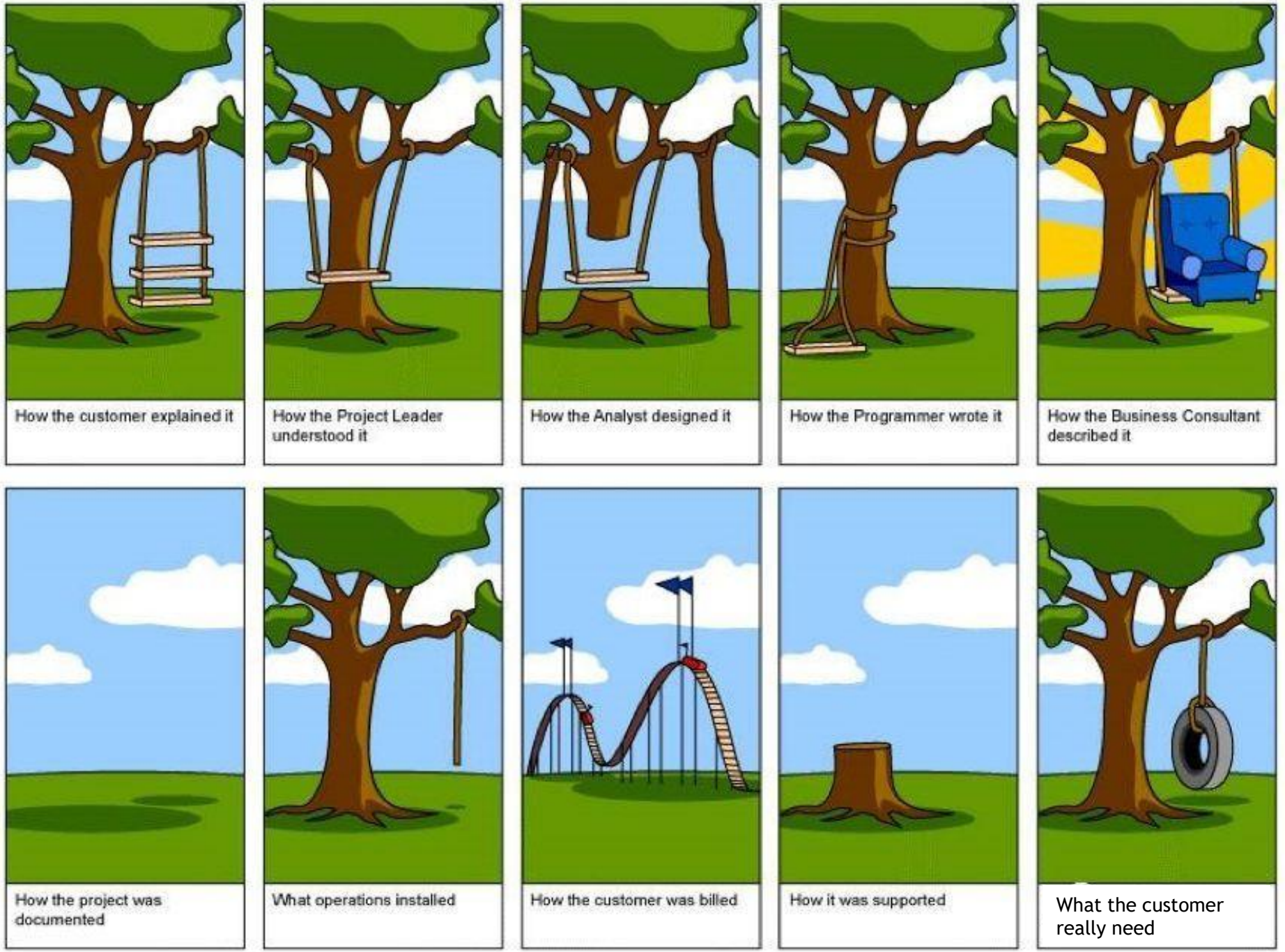
Software Requirements Engineering

Lecture 1: Introduction to Software Requirements Engineering

Outline

1. The importance of software requirements engineering
2. What are software requirements
3. Levels and types of software requirements
4. Software requirement engineering


How the customer explained it
How the Project Leader understood it
How the Analyst designed it
How the Programmer wrote it
How the Business Consultant described it
How the project was documented
What operations installed
How the customer was billed
How it was supported
What the customer really need

Introduction

- Phillip developed a personnel system for a company. One day, he got a call from Maria, who is working in the Human Resources department.

Hi Phil, an employee just changed her name to Sparkle Starlight, and we can't get the system to accept the name change. Can you help?"

She married some guy named Starlight?"



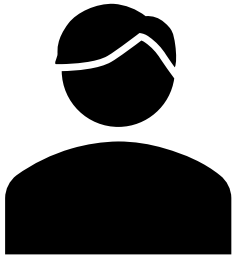
No, she didn't get married, just changed her name. That's the problem. It looks like we can change a name only if someone's marital status changes."



Well, yeah, I never thought someone might just change her name. I don't remember you telling me about this possibility when we talked about the system,

I assumed you knew that people could legally change their name anytime they like. We have to straighten this out by Friday or Sparkle won't be able to cash her paycheck. Can you fix the bug by then?"

Introduction



It’s not a bug! I never knew you needed this capability. I’m busy on the new performance evaluation system. I can probably fix it by the end of the month, but not by Friday. Sorry about that. Next time, tell me these things earlier and please write them down.

What am I supposed to tell Sparkle? She’ll be upset if she can’t cash her check.

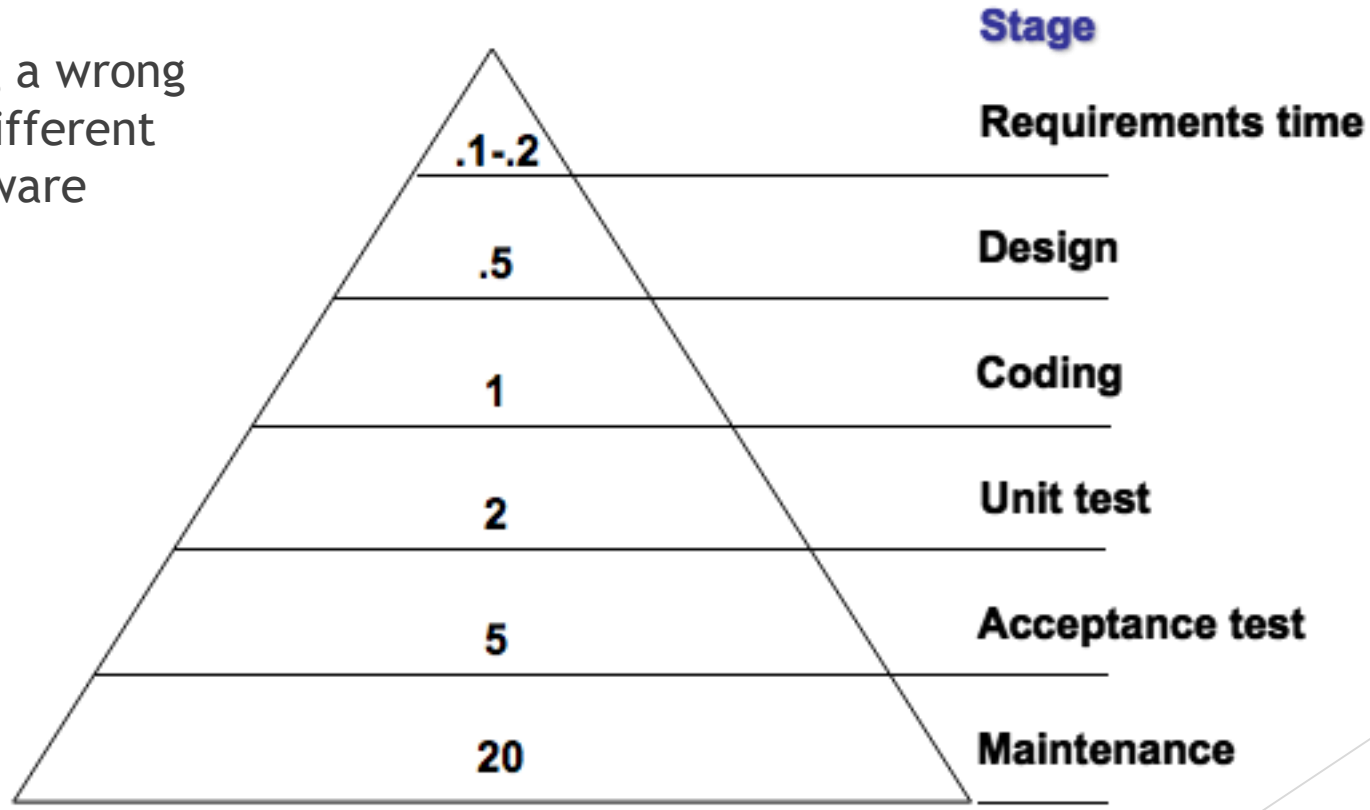


Hey, Maria, it’s not my fault. If you’d told me in the first place that you had to be able to change someone’s name at any time, this wouldn’t have happened. You can’t blame me for not reading your mind.

Yeah, well, this is the kind of thing that makes me hate computers. Call me as soon as you get it fixed

The Importance of the Software Requirements

- The cost of fixing a wrong requirement in different stage of the software lifecycle.



Benefits from a high-quality requirements process

- Fewer defects in requirements and in the delivered product
- Reduced development rework
- Faster development and delivery
- Fewer unnecessary and unused features
- Lower enhancement costs.
- Fewer miscommunications.
- Reduced scope creep.
- Reduced project chaos.
- Higher customer and team member satisfaction.
- Products that do what they’re supposed to do.

What is a requirement?

- It may range from a high-level abstract statement of a service or of a system constraint to a detailed mathematical functional specification.
- This is inevitable as requirements may serve a dual function
  - May be the basis for a bid for a contract - therefore must be open to interpretation;
  - May be the basis for the contract itself - therefore must be defined in detail;
  - Both these statements may be called requirements.

What is a requirement?

- Case study

- Assume that you are asked to design a Calculator, think about how your calculator will look like?
  - What do you want from the Calculator?
  - Why do you need these functionality?

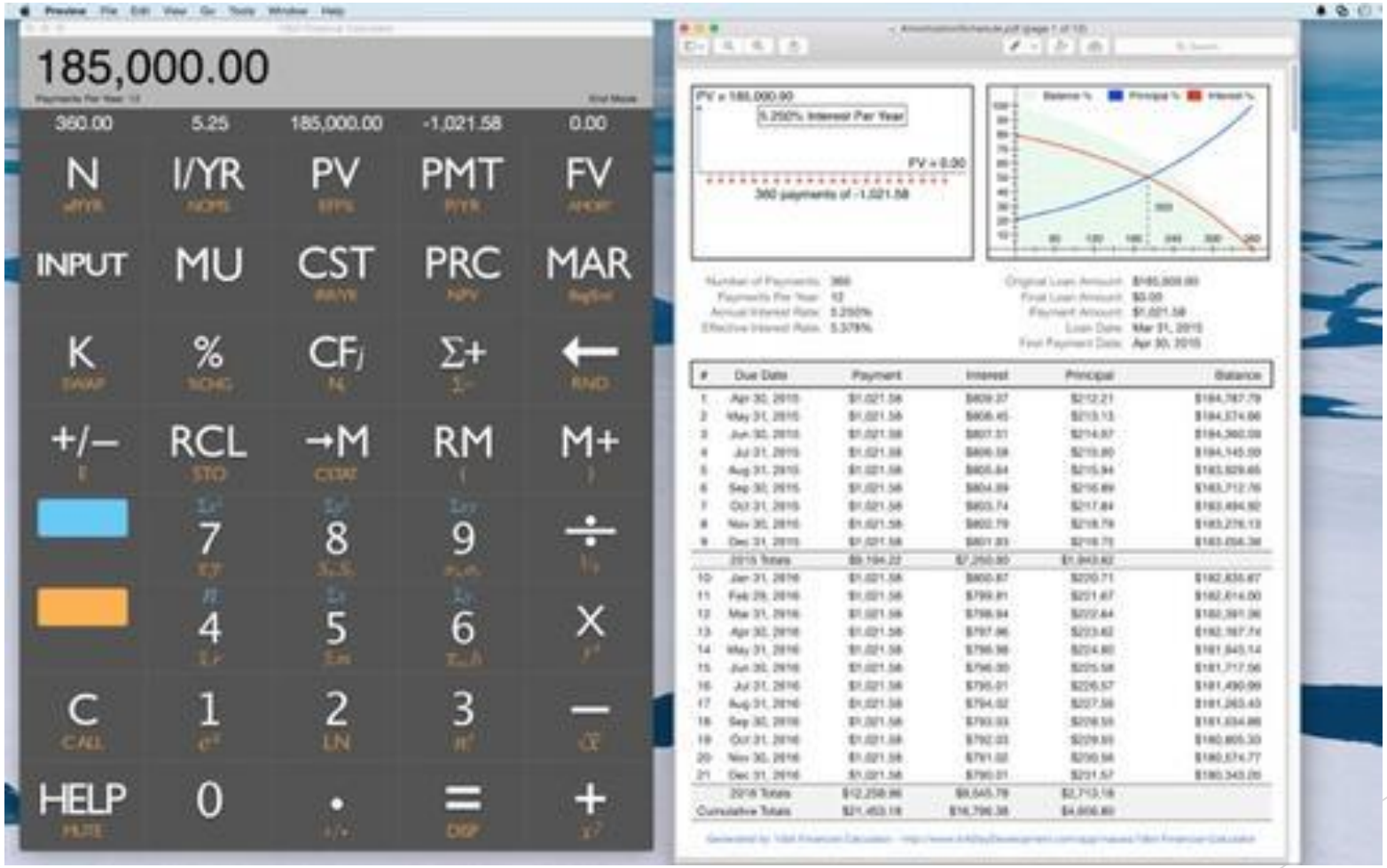
What is a requirement?
Different Calculators







What is a requirement?
Specialized Calculators



What is a requirement?

- Think about the following questions:

- What product is a good product?
- What makes the same kind of products different?
- How are you going to describe your design?

What is software requirement?

- Anything that drives design choices
- A property that a product must have to provide value to a stakeholder

- Not bad, but not precise

- Requirements are a specification of what should be implemented.
- They are descriptions of how the system should behave, or of a system property or attribute. They may be a constraint on the development process of the system.

The Definition of Software Requirement

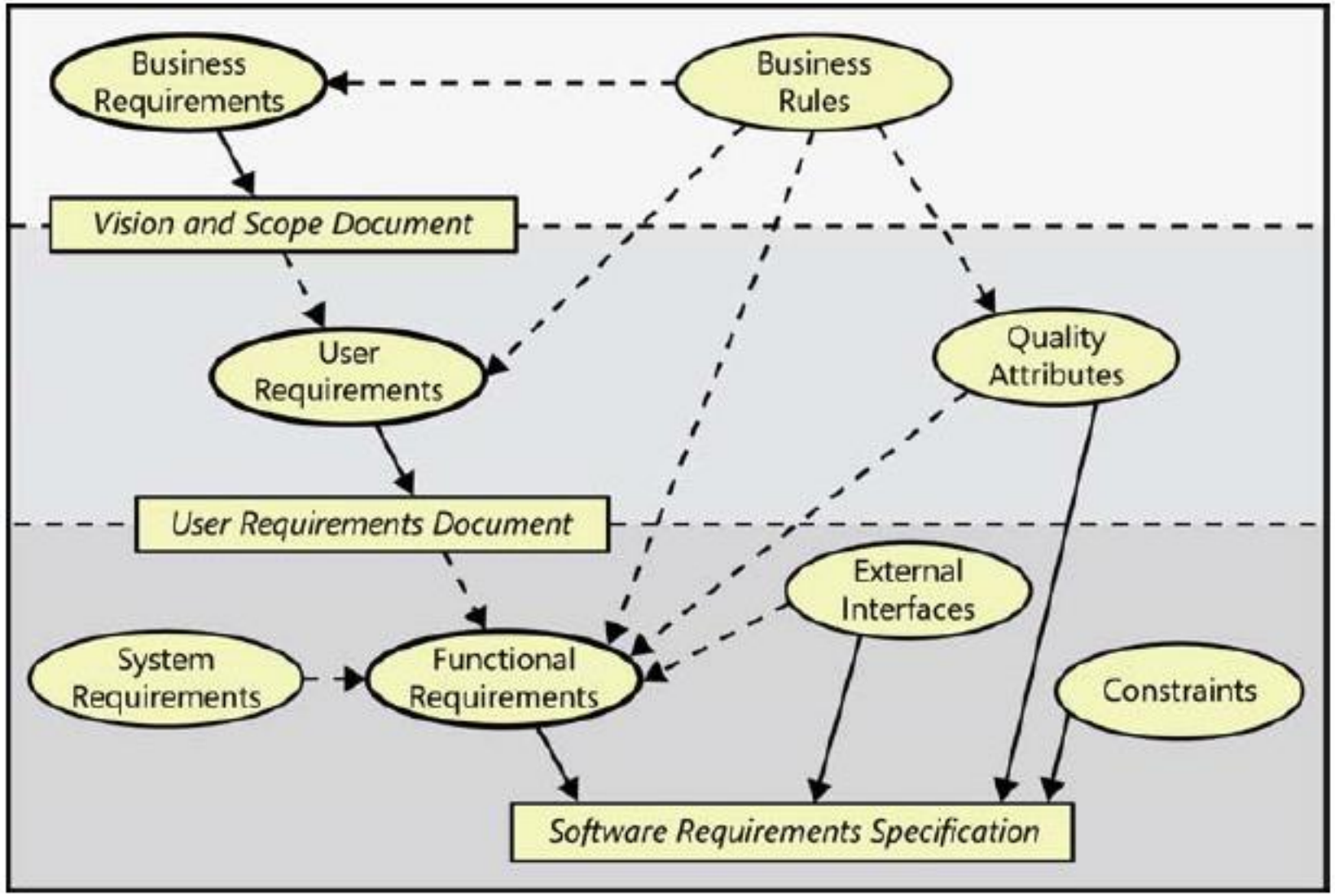
1. A condition of capability needed by a user to solve a problem or achieve an objective
2. A condition or a capability that must be met or possessed by a system to satisfy a contract, standard, specification, or other formally imposed document.

IEEE Definition

What is software requirement?

- What is not a software requirement?
  - Design and implementation details
  - Project schedule information
  - Testing information

Levels and Types of Requirements



Relationships among several types of requirements information.

Solid arrows mean “are stored in”

Dotted arrows mean “are the origin of” or “influence.”

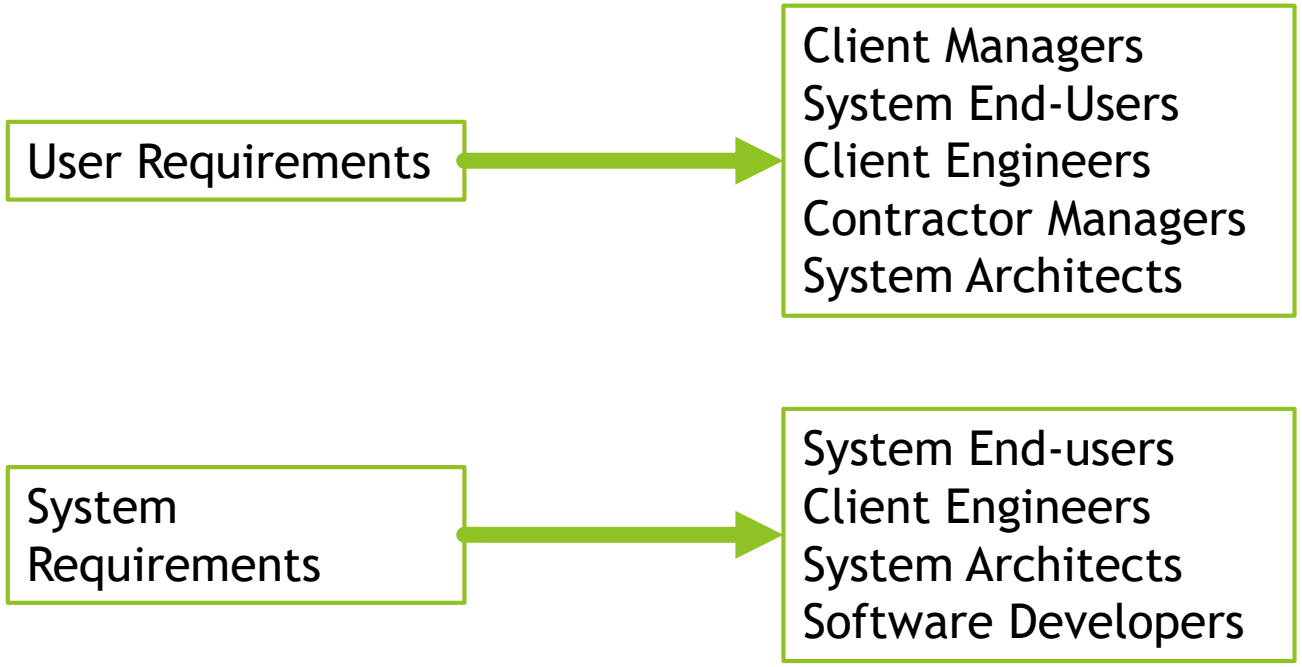
Levels and Types of Requirements

- Business requirements: A high-level business objective of the organization that builds a product or of a customer who procures it.
- User requirements: Statements in a natural language plus diagrams, of what services the system is expected to provide to system users and the constraints under which it must operate.
- System requirements: More detailed descriptions of the software system’s functions, services and operational constrains.

Levels and Types of Requirements

- An example: A mental health care patient management system
- User Requirement Definition
  - The medical center booking system shall generate monthly management reports showing the cost of drugs prescribed by each clinic during that month.
- System Requirements Specification
  - On the last working day of each month, a summary of the drugs prescribed, their costs, and the prescribing clinics shall be generated.
  - The System shall automatically generate the report for printing after 17.30 on the last working day of the month.
  - A report shall be created for each clinic and shall list the individual drugs names, the total number of prescriptions, the number of doses prescribed, and the total cost of the prescribed drugs.

Levels and Types of Requirements

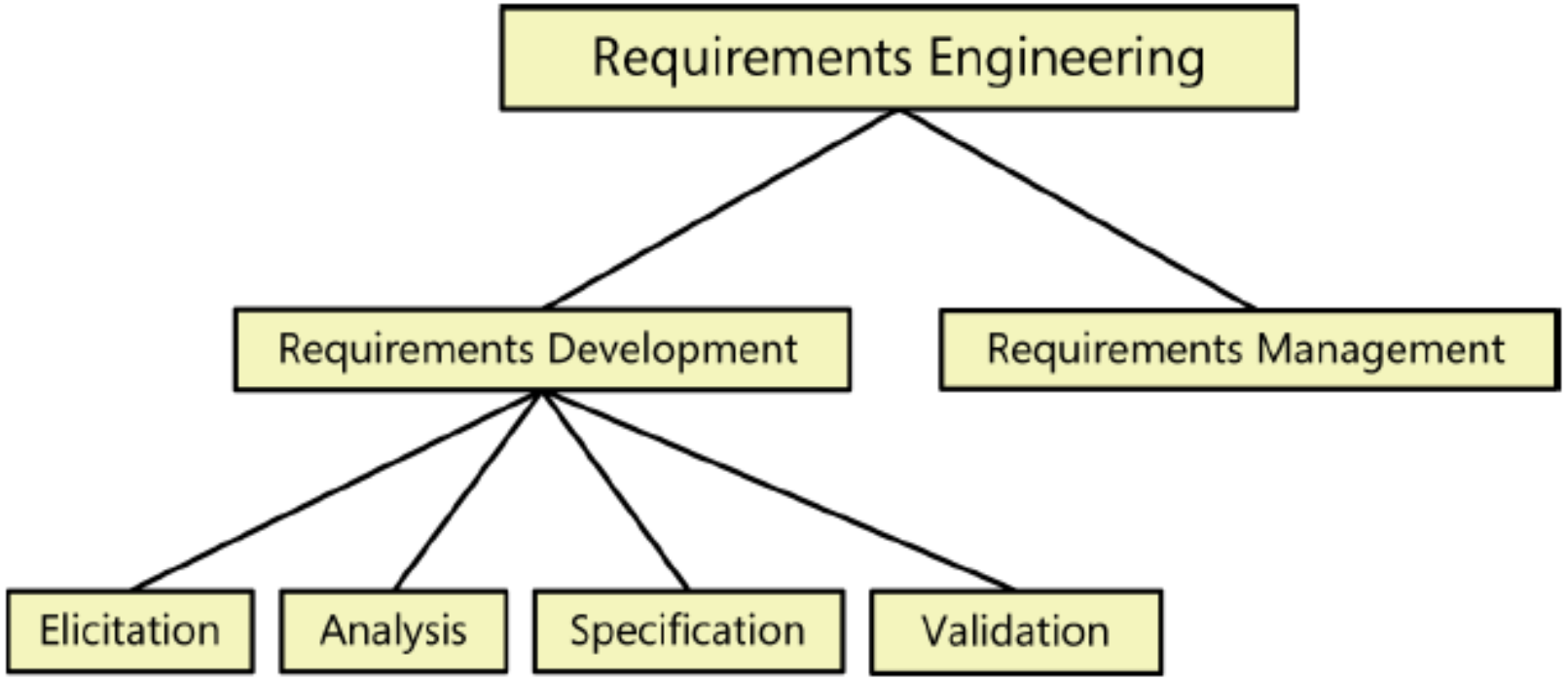


Software Requirement Engineering

Requirements Engineering (RE) is a set of activities concerned with identifying and communicating the purpose of a software-intensive system, and the contexts in which it will be used. Hence, RE acts as the bridge between the real world needs of users, customers, and other constituencies affected by a software system, and the capabilities and opportunities afforded by software intensive technologies…

IEEE standard

Software Requirement Engineering



Requirements Development-Elicitation

- Identifying the product's expected user classes and other stakeholders.
- Understanding user tasks and goals and the business objectives with which those tasks align.
- Learning about the environment in which the new product will be used.
- Working with individuals who represent each user class to understand their functionality needs and their quality expectations.

Requirements Development-Analysis

- Analyzing the information received from users to distinguish their task goals from functional requirements, quality expectations, business rules, suggested solutions, and other information
- Decomposing high-level requirements into an appropriate level of detail
- Deriving functional requirements from other requirements information
- Understanding the relative importance of quality attributes
- Allocating requirements to software components defined in the system architecture
- Negotiating implementation priorities
- Identifying gaps in requirements or unnecessary requirements as they relate to the defined scope

Requirements Development-Specification

- Translating the collected user needs into written requirements and diagrams suitable for comprehension, review, and use by their intended audiences.

Requirements Development-Validation

- Reviewing the documented requirements to correct any problems before the development group accepts them.
- Developing acceptance tests and criteria to confirm that a product based on the requirements would meet customer needs and achieve the business objectives.

Requirements Development

- Iteration is a key to requirements development success.

You're never going to get perfect requirements.

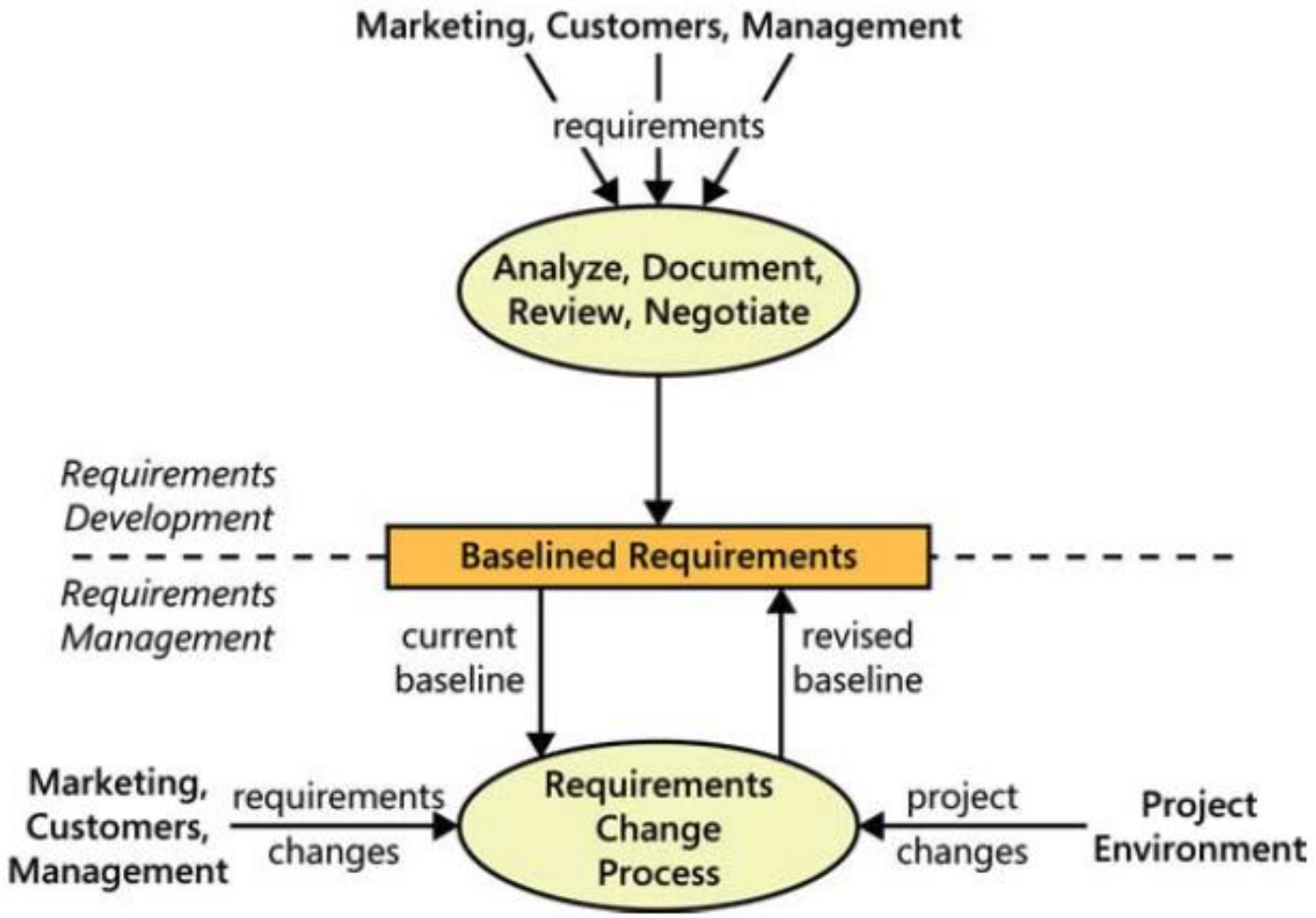
- From a practical point of view, the goal of requirements development is to accumulate a shared understanding of requirements that is good enough to allow construction of the next portion of the product to proceed at an acceptable level of risk。

Requirements management

- Requirements management activities include the following:
  - Defining the requirements baseline
  - Evaluating the impact of proposed requirements changes and incorporating approved changes into the project in a controlled way
  - Keeping project plans current with the requirements as they evolve
  - Negotiating new commitments based on the estimated impact of requirements changes
  - Defining the relationships and dependencies that exist between requirements
  - Tracing individual requirements to their corresponding designs, source code, and tests
  - Tracking requirements status and change activity throughout the project

Requirements management

- The object of requirements management is to anticipate and accommodate the very real changes that you can always expect so as to minimize their disruptive impact on the project.



In the next lecture…

- Review of some software development models
- Requirements engineering and the software development
- Risks in software requirements engineering

Question session